

INTERNATIONAL CONFERENCE ON RECENT TRENDS IN ADVANCED COMPUTING
2019, ICRTAC 2019Identification of Network Intrusion in Network Security by Enabling
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Abstract

Cloud computing is an application of the storage of items used as network services. On their virtual machines, cloud users can access hacker technology that used to emphasize back doors in cloud security. For their ability to replicate hardware environments and share resources, virtual machines are quickly replacing personal computer infrastructures. The hacking and denial of service attacks were revealed to physical devices. On their Virtual Machines, Cloud users can download vulnerable programs, leading to cloud security violations. There are many safety issues that need to be addressed in a partial security environment. There is a data centre in the existing system, where system administrators have full control over the host computers, which can be monitored in a hierarchical way by the system administrator. Patching known cloud safety holes data centre where cloud users are usually entitled to control the software installed on their virtual machines. Essentially it may not operate and may breach the Service Level agreement. We can't find the detection of intrusion. Virtual machines are vulnerable to denial of service attack cloud users can set up susceptible software on their Virtual Machines which actually leads to cloud security violations. Network intrusion detection and countermeasure choice combines intrusion detection processes with attack map analytical procedures. If an attacker targets the server using a user account, the attacker can add multiple levels of malware to the database if and only if he can access the server, but because of the server cloud service it is difficult to detect the intruder in the existing system. In the proposed system, If the attacker's server with the attack user account finds the attacker and administrators to warn the user's attempts (zombie attack) to access another user account to install malware and attempts to manage the largest multi-level wait, then administrators will stop it.

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1. Introduction

Service over a network is delivered by using computing resources of cloud computing. Multi domain and multi user platform is supported by distributed cloud computing. Cloud computing is more prone to security threats as it supports distributed multi user and multi domain platform. Services provided by cloud computing include delivery of applications as service over the internet, the hardware and system software in datacenter. Cloud computing is attracted by many users it provides ubiquitous and pay for use services. Several disadvantages are also included by cloud computing. Security aspects like confidentiality, availability and integrity of cloud are being attacked. Improved cloud security can be achieved by providing security systems to both cloud users and cloud services. Main issue involved in cloud computing is Security. We can detect instructions in a cloud by using different strategies. By analyzing the malicious activity and traffic on the network we can discover unauthorized network access by using network Instruction detection system. The new cloud survey alliance finds the misuse and unethical

use of cloud computing as the main security threat. In a centralized manner system administrator have full control over host machines, vulnerabilities. Service level agreement have been violated as virtual machines (VM) may not function effectively and user do not have privilege to control software installed due to security holes in patching. Loop holes in cloud security are contributed by installing vulnerable software on user's virtual machines. Top concerns of cloud computing systems are "Business continuity and services availability". Legal service contract can be violated if known security holes are fixed in cloud data centers. Essentially right provided to user to control and manage the software installed on virtual machine may not work. Exception in cloud security is contributed by installing vulnerable software on user's virtual machines. Identifying and minimizing the impact of cloud user's security breach is being a very big challenge to administrators.

The infrastructure was shared by millions of users in the cloud system. Using shared infrastructure to exploit the cloud's vulnerabilities and use its resource to more efficiently deploy attacks ways has benefited the attackers. These attacks have been more successful in the cloud environment as users of the cloud typically share computer Resources, such as being linked via the same switch, sharing data storage and file system with potential attackers to compromise multiple Virtual Machine. The system administrators have full control over the host machines in the existing system where the data center is present, system administrator detects vulnerabilities in a centralized manner. Patching known security gaps in cloud data centers where clients of the cloud generally have the right to control installed software on their Virtual Machine. It may not work substantially and may break the agreement on service level. In addition, cloud users can install insecure code on their VMs, which contributes significantly to cloud protection exceptions. The goal is to develop an active vulnerability / attack detection and response system to efficiently identify attacks and minimize the impact on cloud users of security breaches. So it won't work effectively and can break the Service Level Agreement. We cannot find the intrusion detection.

In this article, we propose (identification of Network Intrusion detection in network security) to establish a defense-in-depth Intrusion detection framework. For better detection of attacks, network intrusion incorporates intrusion detection processes of attack graph analytical procedures. Design of network intrusion does not need to improve any of the existing intrusion detection algorithms; Nonetheless, network invasion uses a reconfigurable digital networking strategy to identify and combat Virtual Machine attempts, thus preventing zombie virtual machine.

In common, there are two main phases in network intrusion: (1) Use a network intrusion detection agent (network intrusion -A) to monitor and analyze web traffic on each cloud server. A Network Intrusion, a regularly checks the virtual machine vulnerabilities within a cloud server to create Scenario Attack Map, and based on the magnitude of the found vulnerability to the goals of the collaborative attack, Network Intrusion can determine whether or not to place a Virtual Machine in the Network Inspection State. (2) Once a Virtual Machine is examined, Deep Packet Inspection is implemented and/or virtual network reconfigurations can be introduced to the Virtual Machine inspectorate to improve potential attack behaviors. In the Proposed System, Network intrusion detection and countermeasures option is to establish a framework for better defense in depth IDS attack detection, network intrusion detection and countermeasure selection process includes intrusion detection to attack graph analysis program. When the server is attacked by an attacker by using a user account, Multi-level servers attacker could deploy malicious software, if and only if he can gain access to the server, but difficult to find in the existing system, because the cloud service attack servers.

When attackers attack the attack scene map user account, attack analyzer can detect attacks and send alerts to the administrator (by attacking zombies) attempts to access another user account deploying multilevel malware and using the

maximum number of attempts on a permanent basis, then he waits for the administrator of block management. The Reconfigurable virtual networking was the approach used by the design of network intrusion uses a to detect and count the attempts to compromise Virtual Machine. so preventing zombie Virtual Machine, attack graph-based analytical models and reconfigurable virtual network-based countermeasure is built in the system. Various algorithms were used by them for graph based analysis model like Alert Correlation and Countermeasure Selection. Alert correlation presents Virtual Machine protection model of network intrusion consists of a Virtual Machine profiler, a security indexer, state monitor. Depending on various factors such as connectivity, the number of vulnerabilities present and their impact scores, we specify the security index for all the Virtual Machine in the network. The impact rating of vulnerability as described in the guide helps to assess the impact of the vulnerability being exploited on privacy, honesty and availability. A virtual machine's connectivity metric is determined by assessing incoming and outgoing connections. The counter measure that gives the lowest value of Return Of Investment when applied to a node is called the optimal countermeasure. Finally, Scenario Alert Graph and Alert Correlation Graph will also be updated before the algorithm ends. To calculate which node attack can easily be identified.

2. Related Works

2.1. Cloud Traffic Analysis

Dijiang Huang[5] said that Each cloud server uses a network intrusion detection agent to capture and power form analysis of cloud traffic. The virtual system vulnerabilities are scanned within a cloud server. Intrusion detection alerts are sent by the agent when some suspicious or unprotected traffic is found for the purpose of attack analysis. by use of methods like SPAN, RSPAN, or ERSPAN, the traffic generated on the mirrored software bridge by the virtual machines will be mirrored to a particular port on a particular bridge. The sniffing rules of intrusion detector can be customized based on our needs. Dom0, a privilege domain in the Xen environment which consists a virtual switch for traffic switching among Virtual Machine and network drivers for physical network interface of the cloud server.

2.2 Virtual Machine Configuring

B.Joshi&co[3] said that, is possible to profile virtual machines to obtain accurate state information, running services, open ports, etc. The communication with other Virtual Machine is a major factor that counts towards a Virtual Machine profile. An Virtual Machine connected to many other machines is more vulnerable than one that is connected to fewer machines end to cause more damage due to the compromising effect of a highly connected Virtual Machine. To verify the authenticity of alerts related to a Virtual Machine, knowledge about the services running on a Virtual Machine is essential. An attacker deploys a port scanning program in order to conduct an intensive network review to find out which ports are open on any Virtual Machine. Vulnerability of a virtual machine is determined based on the information on any Virtual Machine open ports and the history of open ports. All these factors together the virtual machine profile will be created. Virtual machine profiles are stored in a database and contain comprehensive vulnerability, warning, and traffic information. Attacks are analyzes of different data.

2.3. Attack Analysis:

RadharaniSajja[1] discussed that an attack analyzer performs operations update, warning correlation and countermeasure collection, for an example, The Scenario Attack Graph function made up of three phases such as information gathering, potential exploit path analysis and attack graph construction. With these information, the attack paths are modelled using Scenario Attack Graph. When attack analyzer receives an alert from a network intrusion -A, it matches that alert in the Alert Correlation Graph. If the received alert appears in the graph, the implementation of response analyzer selection procedure and informs the network controller actions deploy these countermeasures..

2.4. Security Analysis

To construct attack graphs, it is important to collect network information and provide the same as input to the attack analyzer. The controller of the network is also responsible for deploying the countermeasure. Based on the best return on investment and risk probability nodes, anti-attack analysis, and select executed by the network controller.

2.5.Alert Correlation

Dijiang Huang &co [5] presents Virtual Machine protection model of network intrusion which consists of a Virtual Machine profiler, a security indexer, and a state monitor. A security index is provided for all the Virtual Machine in the network based on different factors such as connectivity, the number of vulnerabilities and their effect ranking. The vulnerabilities affect the score, defined by a guide to help determine the confidentiality, integrity and availability impact of vulnerabilities exploitation. Connectivity is a measure of the virtual machine's incoming and outgoing connections made by the evaluation. Virtual machine state is defined as follows 1. Stable is one of the known vulnerabilities on the virtual machine. 2. Fragile is one or more of those vulnerabilities in a virtual machine, which is still untapped 3. Exploited-the manipulation of at least one flaw, and the virtual machine is conflicted existence. 4. Zombies is a virtual machine is in control of the attacker.

2.6.Counter measure Selection Algorithm

Y. Dong, Dijiang Huang [5][6] presents how to select the optimal counter-measure for the scenario of an attack. The algorithm input is a warning, an attack graph G , and a pool of counter measures. The algorithm starts with the node choice Virtual Alert that corresponds to the alert generated by a network intrusion -A. We calculate the distance of the target before the alert virtual node selection strategy. If the distance is greater than the threshold, we do not choose countermeasures, but the update tracking alarm system alert correlation graph. Virtual alert source node, all reachable nodes (including the source node) is collected into action after the assets of T. Since the attacker has performed before an alarm is generated, the probability of our virtual alert is set to 1, calculate the new probability in the set T all children (downstream) node. Now, the probability of being selected and the new calculated based on the effectiveness of the chosen measures for all $t \in T$ applicable. For the application of measures to deal with the use of the benefit is given by the change in the probability of the target node. In the next double for the cycle, we calculate on the basis of the investment, the rate of return applications welfare measures (ROI). Wherein, when a counter measure is applied at the node so that the minimum value of the ROI, is considered a best countermeasure. Finally, scenarios attack graphs and alarm correlation diagram also terminated before updating algorithms. The complexity of the algorithm 2 is $O(|P| \times |X| \times |DM|)$, where $|P|$ is the number and vulnerabilities $|DM|$ represents the number of responses.

3. SIGNIFICANCE OF DESIGN

Nowadays, in order to service the providers and customers, researches was carried out by the cloud computing process. There are many security issues to be taken care in a cloud computing environment. VM is susceptible to denial of service attack. Cloud user can set up susceptible program in their VM that really conduces a breach in cloud protection. To stop negotiating vulnerable Virtual Machines in the cloud, Intrusion Detection and Countermeasure in virtual Network systems technique has been proposed in this paper. Detection of network Intrusion and counter measure choice in virtual network systems provides a basis for detection of defense-in-depth intrusion. The attack graph analytical procedures were incorporated into the processes of intrusion detection to better detect attack by the virtual machine. A data center exists in the existing system, where system administrators have complete host computer control, system administrator can identify bugs centrally. Patching known safety holes in cloud data centers where cloud users are in general entitled to access installed software on their Virtual Machine. It might not really work. Essentially it may not operate and may impact the SLA. Installation of insecure code by cloud users on their virtual machine, makes significant difference to exceptions to cloud protection. This minimizes the effect on cloud users of the security breach. When an attacker plans to attack the user in a network, it is externally difficult to detect the affected server because cloud users can install many software types with their server user account. It will create every possible paths of attack. So scalability is the big problem for this solution. Server quality is poor since insecure software can be mounted on their virtual machines by cloud users. It offers low security.

3.1.System Design

The framework for network intrusion detection within a cluster of cloud servers is shown in Figure 1. Major components of this device are distributed on cloud server, a network controller and an attack analyzer. The latter three elements are in a centralized setup on each cloud server, connected to computer switches. Network Intrusion Detection is a computer agent that is realized in server which is linked to the control centre through a secured channel isolated

from the usual data packets using Open Flow tunnelling or Virtual LAN approaches. It is the duty of the network operator to implement counteractions in line with the decisions of the attack analyzer.

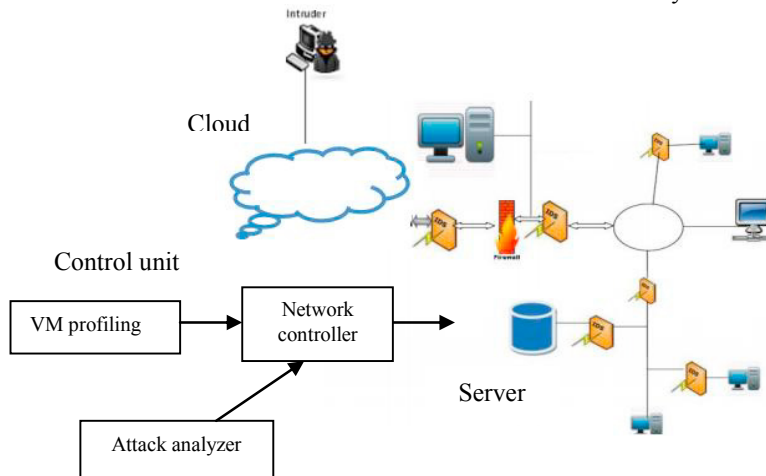


Fig.1. NETWORK INTRUSION ARCHITECTURE

3.1.2. Design And Implementation of Proposed System

A key is used by us to evaluate whether or not the attack order originated from the main attacker is shown in Figure 1. The compromised Virtual Machine checks the key when the attack order is received. When the key is getting matched, the attack is coordinated and initiated. When the match is not successful, the VM must sit tight. We are responsible for communicating among the main attacker, the zombie computers, and the victim or target server. We have a connection between the above mentioned counterparts to establish a communication. If it detects multiple requests from one or more IP addresses, those IP addresses will be placed in the blocked list for a certain period of time. Requests are free to be sent again by these IP addresses after that time span expires. However, if they throttle the server again, three warnings can be made and the admin can finally block the page. In this article, a model for the detection of security intrusion is proposed for network intrusion detection and countermeasure choice within virtual network systems. For better identification of attacks, attack graph analytical procedures are incorporated into the processes of detection of intrusion by network intrusion. Improvement of any of the current algorithms for intrusion detection is not intended by network intrusion design and should be noted by us. Virtual Machine attempts are detected in fact and countered by a reconfigurable virtual networking approach that is employed by network intrusion. When the intruder uses a user account to attack the server. The attacker deploy many levels of malware to the server if the server can be accessed by the attacker, but it is difficult to find the attacker in the existing system due to the cloud service. But in proposed, when the server is attacked by the attacker by using a user account, the attacker can be detected by the attack analyzer and the warning is sent to administrator that the multiple levels of malware are deployed through access to other user accounts which is tried by user attacked by the zombie. Admin requires full attempts and then admin uses scenario attack graph to permanently block him.

The attack is analyzed by the various algorithms used, such as Alert correlation. It can predict potential threats and attacks by correlating detected events or activities. If the event is identified as future attack, particular counter steps can be applied by it to its impact can be reduced or action can be taken In order to prevent damage to the cloud system and counter measure used to reconfigure and monitor the virtual network based system.

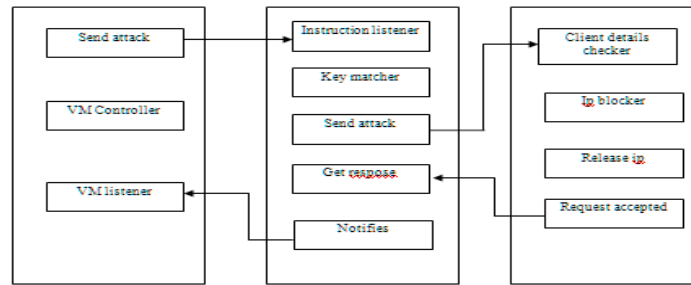


FIG 2: IMPLEMENTATION OF PROPOSED SYSTEM

3.1.3. Summary

A computer security network detection system for intrusion is an device designed to detect unwanted entry to a computer network by monitoring network traffic for traces of unwanted activity. A recent Cloud Study Alliance survey shows that misuse and unethical access of cloud computing are the main security. Bugs can be found and repaired centrally by the system administrator in modern data centres where the complete control is taken care by administrators of the host machines. Cloud users can also install insecure code on their Virtual Machine, which greatly leads to vulnerabilities in cloud protection. Now, in terms of service providers or consumers, there is an emerging research day in cloud computing. There are many security issues that need to be addressed in a cloud computing world. In order to avoid prone virtual machines being negotiated in the cloud, we are projecting an countermeasure in virtual Network systems technique and intrusion detection in this paper. Detection and countermeasure selection of network intrusion in virtual network systems establishes a framework for detection of intrusion in defence-in-depth it incorporates analytical methods of attack graph in intrusion detection Processes for better detection of attacks. There is a data center in the existing system, where System administrators have complete host computer control, system administrator may find bugs centrally. In this paper, we propose detection of Network Intrusion and selection of countermeasures in virtual network systems to create a structure for detection of defence-in-depth intrusion. Network intrusion integrates attack graph analysis techniques in intrusion detection systems for better detection of attacks. It should be noted that the development of network intrusion does not intend to improve any of the current algorithms for intrusion detection; indeed, network intrusion uses a reconfigurable virtual networking approach to identify and combat attempts to compromise Virtual Machine, thereby preventing Virtual Machine zombie. The proposed system incorporates network intrusion detection and countermeasure selection as a defence-in- depth intrusion detection method for better detection of attack. The attack analyzer detects the attacker when an intruder targets the server using the user account and gives the administrator an alert that the client (attacked by the zombie) attempts to access another user account to execute many levels of malware and admin requires for full attempts and then administrates it permanently using scenario attack map. If incident is defined as a potential attack, numerous counter measures can be introduced to mitigate its impact or take action to prevent damage to the cloud system and countermeasure to reconfigure and monitor the electronic network-based system. Countermeasure is a process, operation, program or device capable of preventing or minimizing the impact of attacks on a network computer. Countermeasure is chosen by an attack analyzer and performed a network controller.

4. EXPERIMENTAL RESULT

In this paper, we addressed network intrusion, where mixed attacks in the cloud virtual Networking system are suggested to be detected and mitigated. Intrusion of the network uses the design of the attack graph to detect and predict attacks. The solution proposed discusses how computerbased methods can be used to improve efficiency of detection. This prevents stages of coordinated assaults on victim manipulation. The analysis of system performance illustrates the viability of network penetration and indicates that the proposed solution greatly reduces the

risk of internal and external attackers abusing and abusing the cloud infrastructure.

Network intrusion examines the network IDS strategy only to combat exploratory attacks by zombies. To increase detection reliability, host based IDS systems need to be integrated to cover the entire range of IDS in the environment. In addition, as suggested in the paper, the extrapolation of the proposed network intrusion approach will be examined by examining the distributed network control. Figure 3 demonstrates the visualization of the Virtual Security Index before countermeasure choice for these virtual machines and where Virtual Machine in the private network is the object of the attackers. Another test environment was created by extending the configuration to create another test environment that includes 14 VMs across 3 cloud servers for performance analysis and capacity testing. To build dedicated SAG for virtual machine, it configured each Virtual Machine as a target node. These Virtual Machines consist of Windows and Linux (L) machines in the 172.16.11.0/24 private network and contain more vulnerability related to their operating system and app. Figure 4 compares the values of the Virtual Security Index before and after the countermeasure is applied, i.e. filtering rules are established. Upon applying countermeasure to all Virtual Machine M4, it indicates the percentage change in the Virtual Security Index. Use counter measures to avoid vulnerabilities and cause the Virtual Security Index to drop without blocking normal services.



Fig. 3. VM Security index

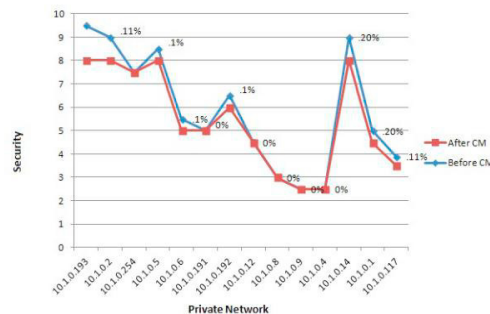


Fig.4. Change in VM security index

4.1. After Analysis of Network Intrusion:

Under various network intrusion, the contact lag with the network is shown in Figure 5. To test the delay, we produced standard packets compared to Network Intrusion -A running in DomU and Network Intrusion running in Dom0. In the comparative study, we record the minimum, average, and peak contact delay. Results demonstrate that the proxy-based Network Intrusion-A lag is the highest because it has to be transmitted by every packet. DomU's Mirror-based Network Intrusion-A and Dom0's network intrusion-A have no significant delay variations. In summary, the delay performance of the Network Intrusion-A at Demo0 and Mirror-based Network Intrusion-A at DomU is better.

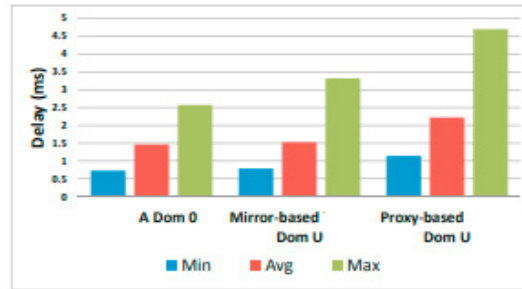


Fig. 5. Network Communication Delay of Network intrusion

5. CONCLUSION

In this paper, we addressed network invasion, which is proposed to identify and mitigate collaborative attacks in the cloud virtual networking environment. Network intrusion uses the design of the attack graph to detect and predict attacks. The approach suggested investigates how to use programming approaches focused on computer switches to increase accuracy of detection and conquer victim exploitation phases of collaborative attacks. The system performance analysis illustrates the viability of network interference and indicates that the proposed solution will substantially reduce the risk of internal and external attackers abusing and abusing the cloud infrastructure.

6. FUTURE WORK

Network intrusion only examines the IDS strategy of the network to combat explorative zombie attacks. To improve detection accuracy, it is necessary to incorporate host-based IDS solutions and wrap the entire spectrum of IDS in the cloud system. For future work, this should be investigated. In addition, as indicated in the paper, through investigation According to the current model Distributed Network Control and attacks, we will investigate the proposed network intrusion prevention solutions scalability.

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